



Medical Microbiology

Third stage- College of Dentistry

Lab.6

Staphylococci

Prepared by

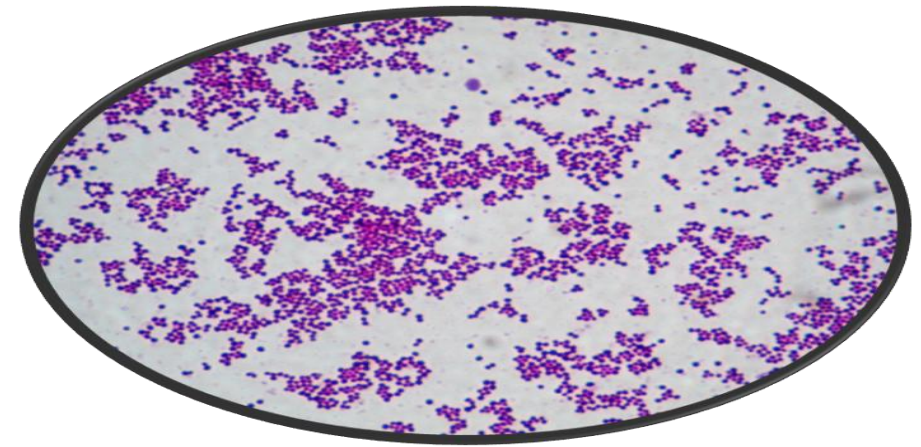
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Learning Objectives:

After this lab. You must be able to:

- 1- Describe staphylococci under microscope.**
- 2-Distinguish between staphylococci and streptococci.**
- 3-List diseases caused by some spp. of staphylococci.**



I. General Characteristics & Microscopy

-Genus: Staphylococcus

-Gram Stain Reaction: Gram-positive (stain purple/blue) due to a thick peptidoglycan layer in the cell wall.

-Morphology: Cocci (spherical or round-shaped bacteria).

-Arrangement: Characteristically arranged in irregular, grape-like clusters.

-Size: Approximately (0.5 - 1.5) μm in diameter.

-Other Features:

Non-motile.

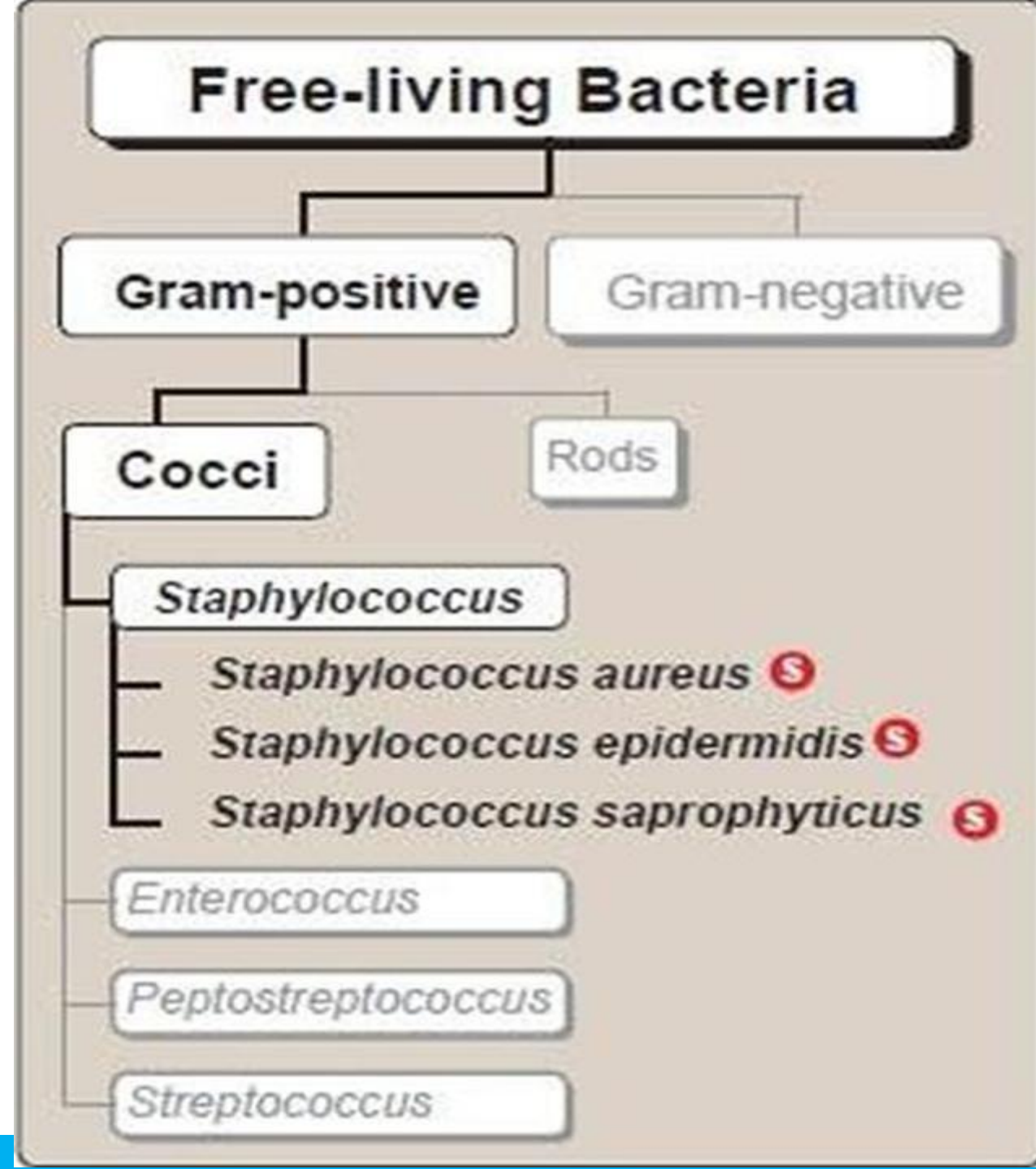
Non-spore forming.

Usually non-encapsulated (though some strains may be capsulated).



-Also aerobic and facultative anaerobic, grow readily on usual culture media, has large, raised, opaque colonies with smooth entire margin.

-Salt tolerant: allows them to tolerate the salt present on human skin.



Clinically important species:

- Staphylococcus aureus*: most virulent species. It is responsible for a wide range of medical illnesses extending from mild localized skin infection to life threatening septicemia.
- Staphylococcus epidermidis*: it is normal flora of the skin and mucous membranes, it is not usually pathogenic but it may cause serious infections.
- Staphylococcus saprophyticus*: it is free living, common cause of UTI in women, and urethritis in males.

S. aureus

Skin infections

Osteomyelitis

Food poisoning

Toxic shock syndrome

Pneumonia

Acute endocarditis

S. epidermidis

Prosthetic Infection

UTI especially in patient
with catheter & low
immunity.

S. saprophyticus

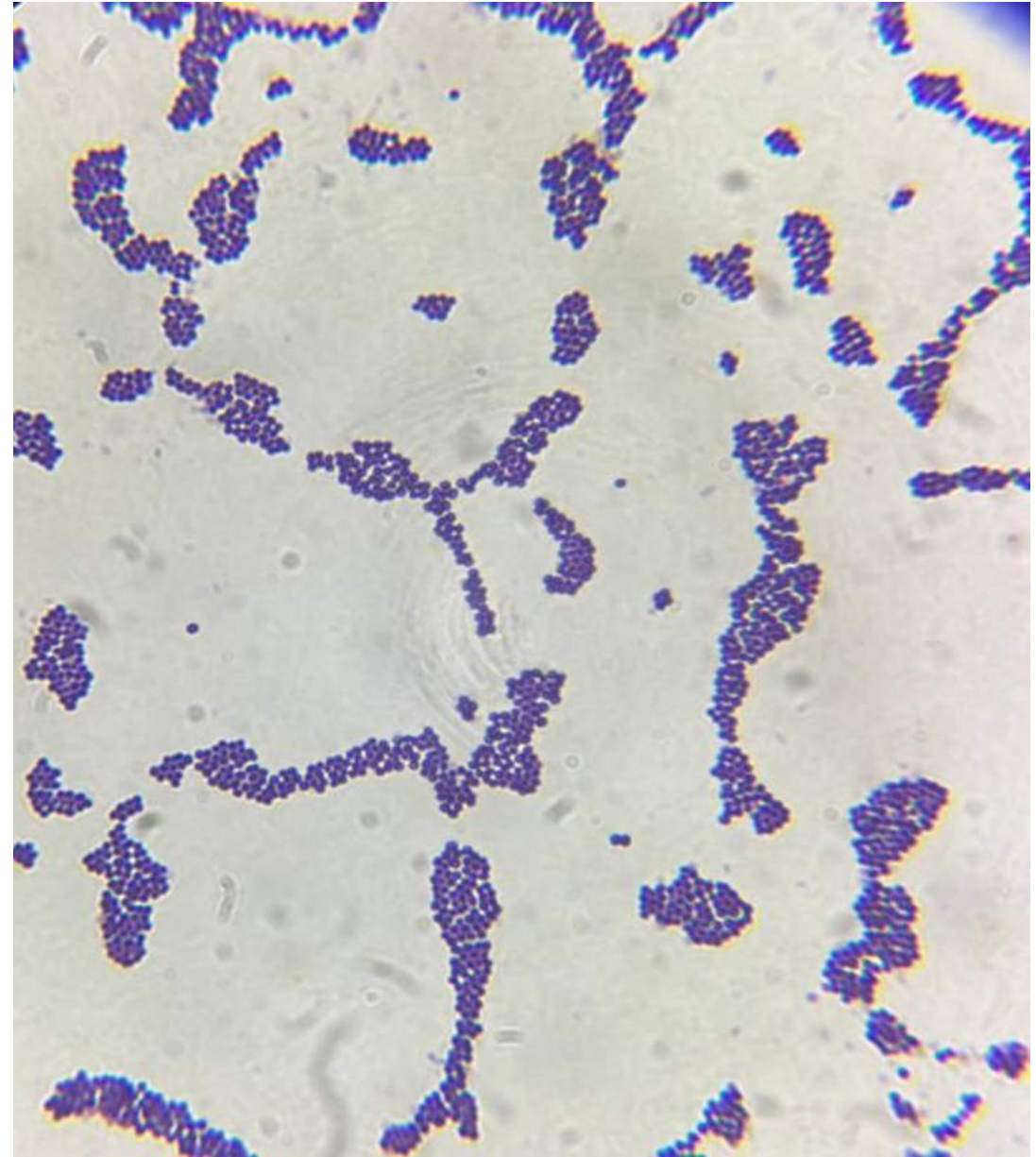
UTI in sexually
active females

Non Gonococcal
Urethritis in males.

Isolation and identification of **Staphylococci**:

Sample:

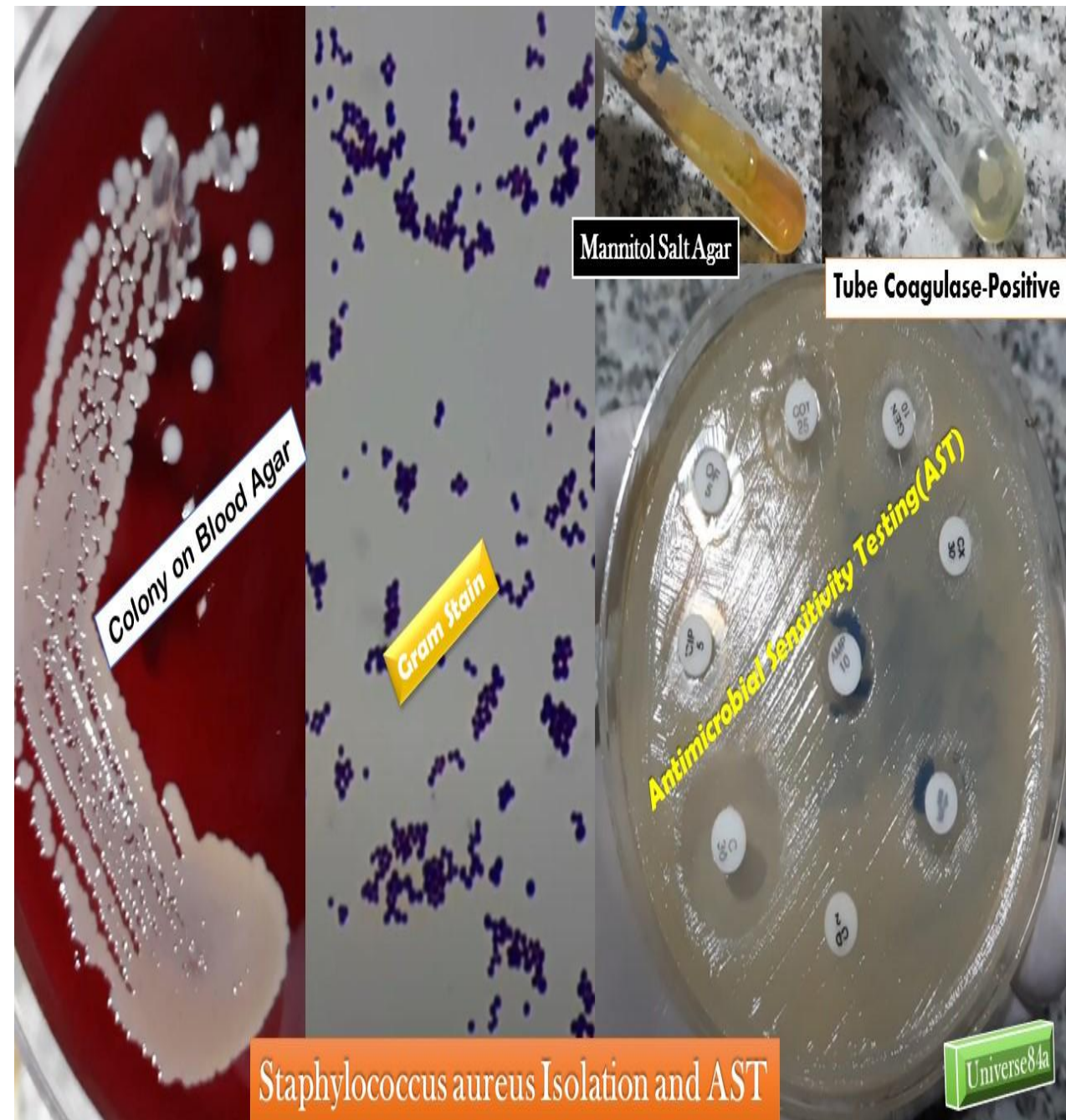
- 1- blood.
- 2- urine.
- 3-stool.
- 4-CSF.
- 5-PUS.



Laboratory identification

-Identification of an isolate as a staphylococcus relies largely on microscopic and colony morphology and catalase positivity.

-*S. aureus* is distinguished from other species coagulase negative staphylococci primarily by the coagulase positivity.



Biochemical tests

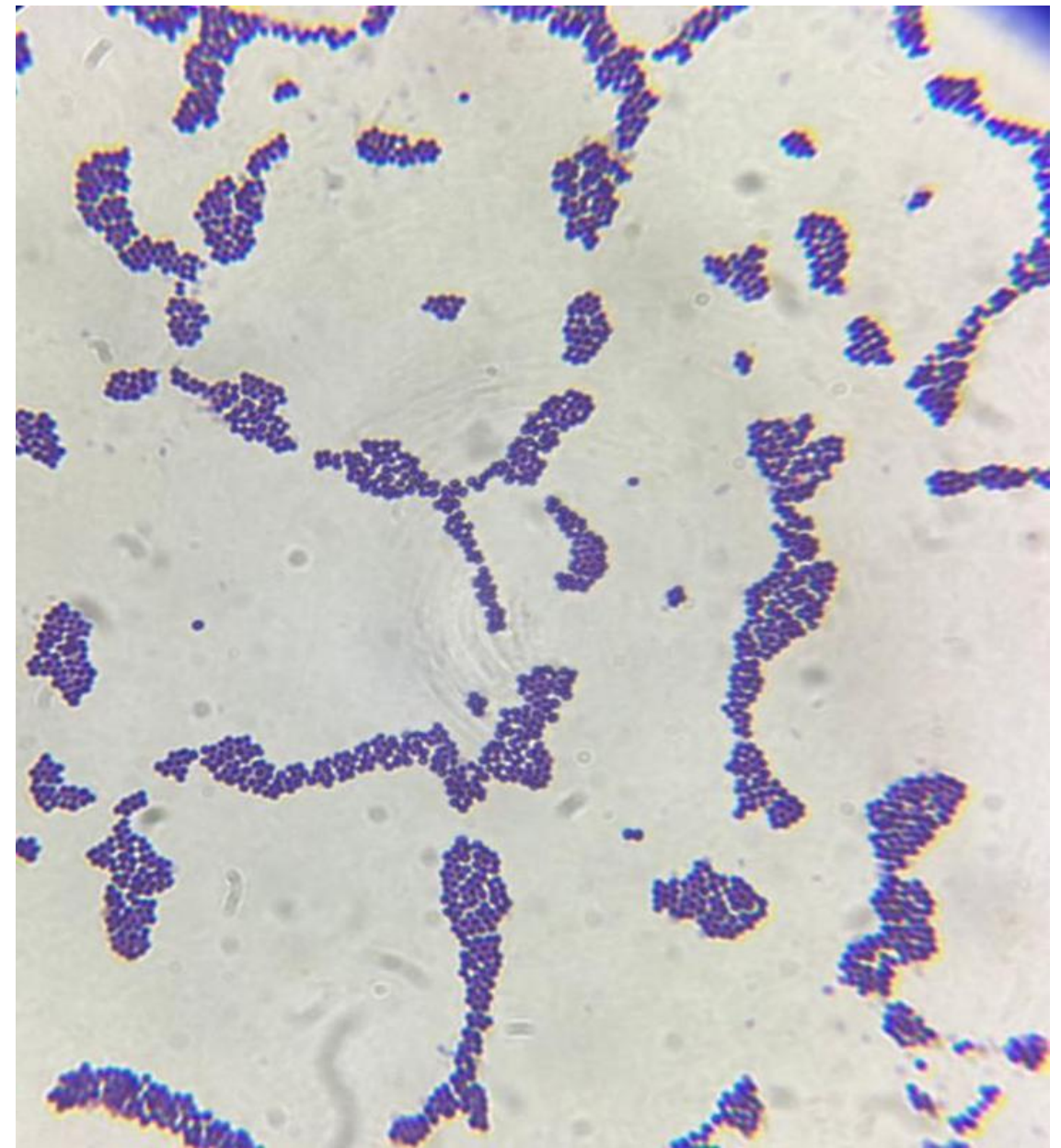
1- Gram stain

When examining the *Staphylococcus* smear under the oil immersion lens, you should observe:

Color: Deep purple/blue (confirms Gram-positive).

Shape: Spherical cells.

Arrangement: Cells grouped together in irregular, clustered arrangements.



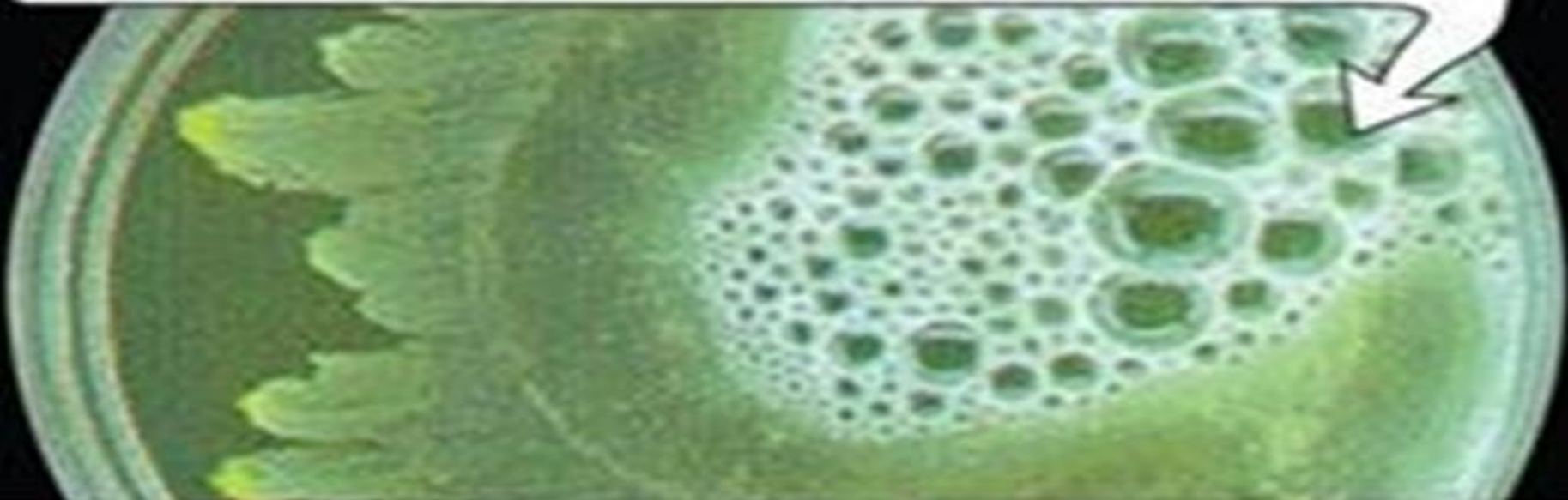
2-Catalase test:

Although microscopic examinations of stained smears between Staphylococci and streptococci, a definitive classification can be made on the basis of the presence or absence of the enzyme **catalase**. Staphylococci contain this enzyme, streptococci do not.



Catalase test

The test for catalase is performed by removing a colony to a slide with a drop of 3% hydrogen peroxide. Catalase-positive cultures produce O_2 bubbles. Here it is demonstrated more dramatically directly on a plate.

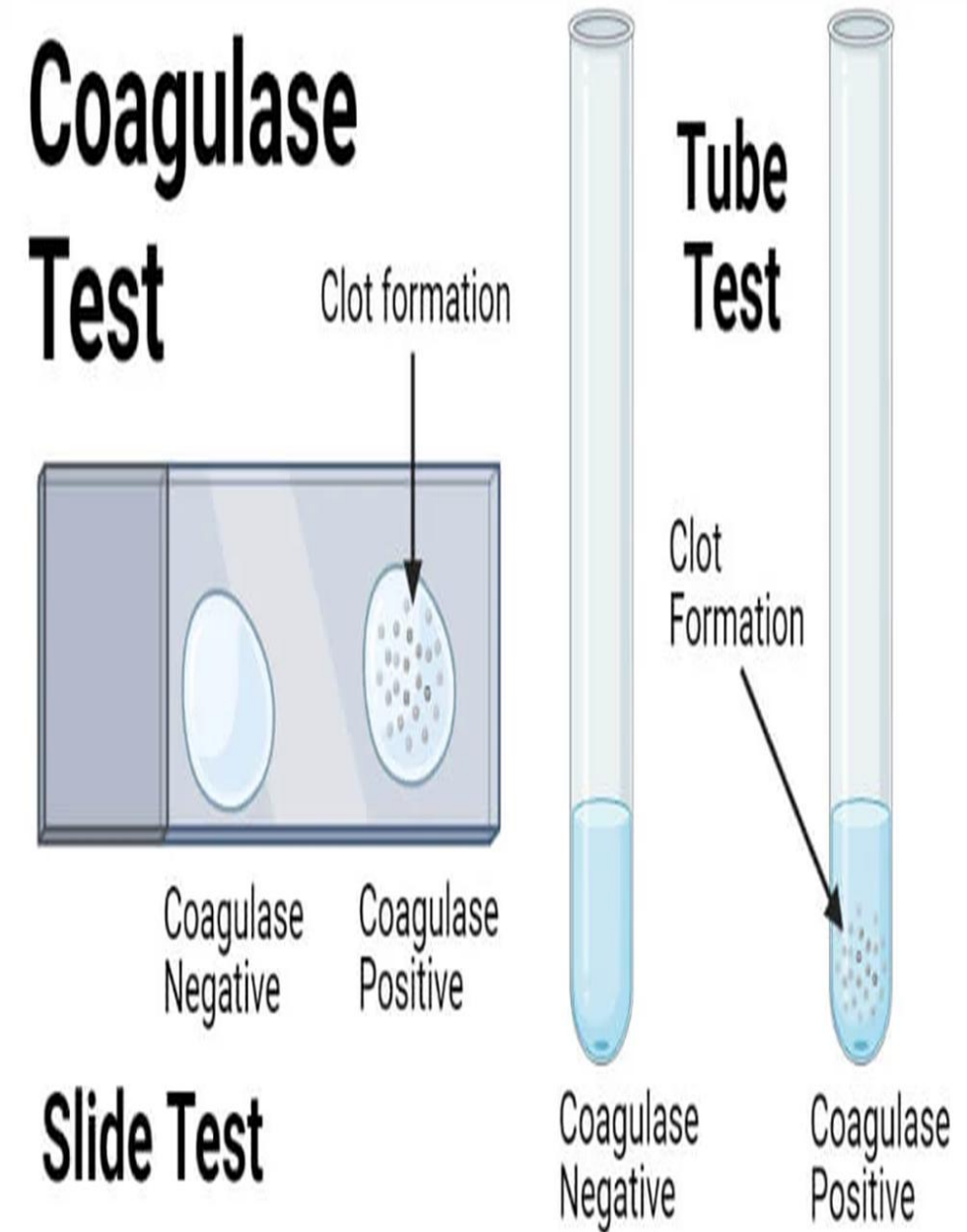


3- Coagulase Test (Differentiation of *S. aureus*):

Purpose: To differentiate the highly pathogenic *S. aureus* from other staphylococci.

Result for :Positive (Plasma clots - forms a solid fibrin gel).

Result for Coagulase-Negative Staphylococci, like: *S. epidermidis*. (No clot formation).



4- Mannitol fermentation on Mannitol Salt agar (MSA):

Staphylococci are able to tolerate the high salt concentration found in Mannitol Salt agar and thus grow readily. If mannitol is fermented, the acid produced turns the phenol red pH indicator from red (alkaline) to yellow (acid).

Procedure:

Mannitol Salt Agar Test

The medium contains **7.5% NaCl, mannitol, phenol red, and peptone and beef extracts**. If the test bacteria are able to grow on this medium, it indicates their tolerance to the high salt concentration.

Result Interpretation:

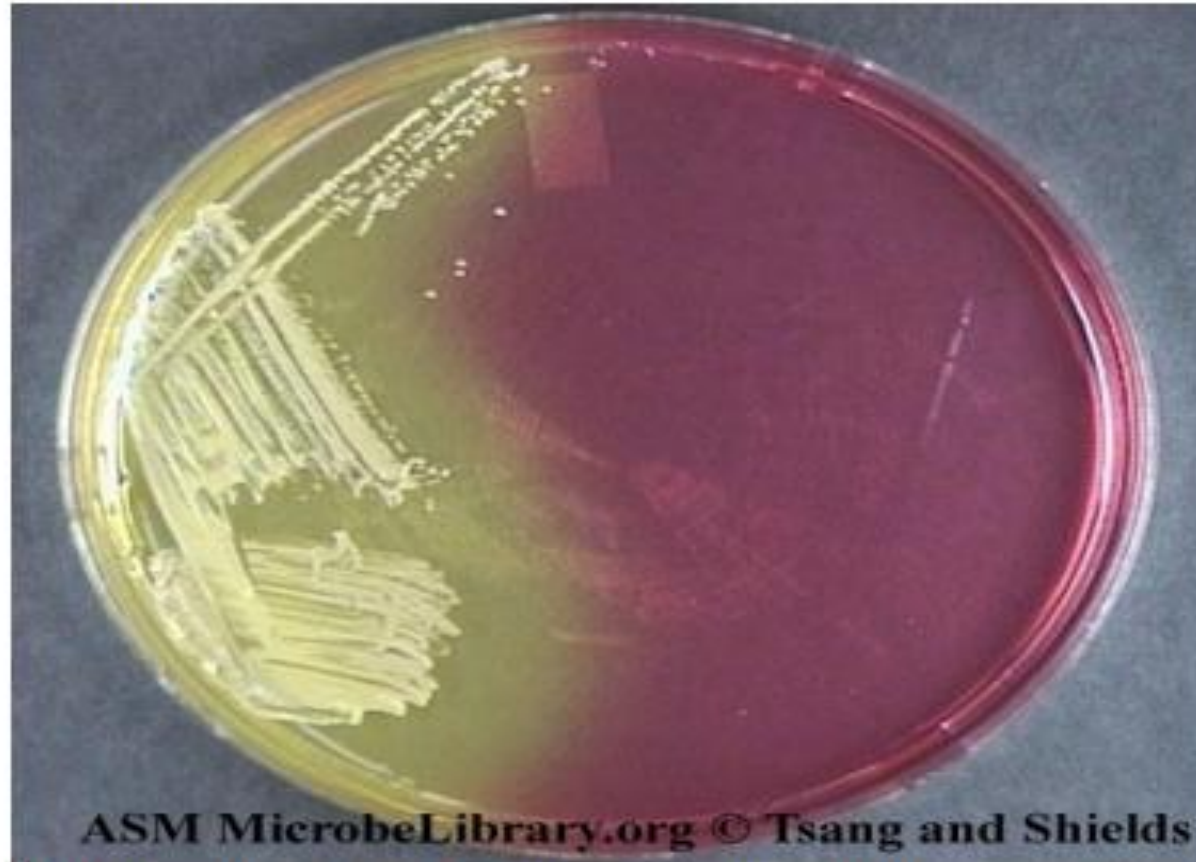
-Positive test result : *S. aureus*, Ferments mannitol, leading to yellow colonies and a yellow color change in the surrounding medium.

- Negative test result : *S. epidermis*, Does not ferment mannitol, resulting in red/pink colonies and no color change in the medium.

Test	<i>S. aureus</i>	<i>S. epidermidis</i>	<i>S. saprophyticus</i>
Mannitol fermentation	Positive	Negative	Usually positive

Positive = acid end products turn the phenol red pH indicator from red to yellow

negative = phenol red remains red



Staphylococcus aureus turning the indicator from red to yellow



Any Question



Thank you